RSA model of referential expression production and comprehension under noise

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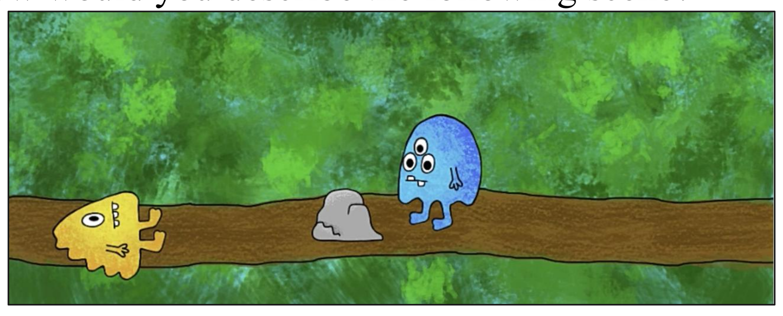
Monster stories

- How would you describe the following scene?

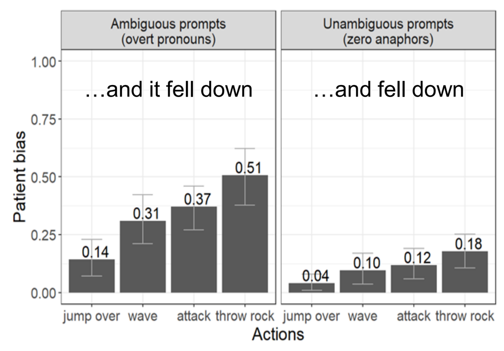


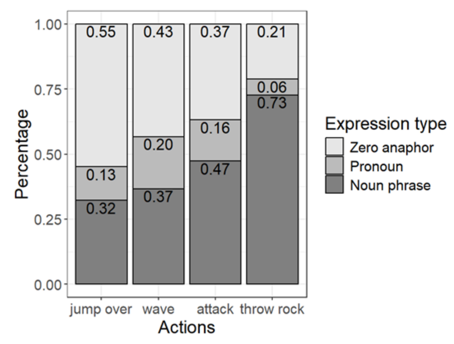
The yellow monster threw a rock at the blue monster and
  a) *... it fell down*
  b) *... fell down*
  c) *... the yellow monster fell down*

- a) is ambiguous and can be easily misinterpreted towards a more plausible event (the blue monster fell).
- b) is not ambiguous but can be confused with a) in an imperfect communication channel.
- c) is the best choice when the event is surprising.

Experimental findings[1,2]

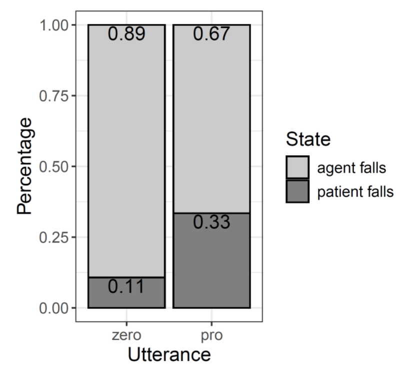
- Listeners, under noisy conditions:
  - interpret ambiguous pronouns according to their prior expectations
  - mentally insert pronouns instead of zero anaphors if it allows them to arrive at a more plausible interpretation
- Speakers, when describing surprising events, use:
  - more noun phrases
  - fewer zero anaphors

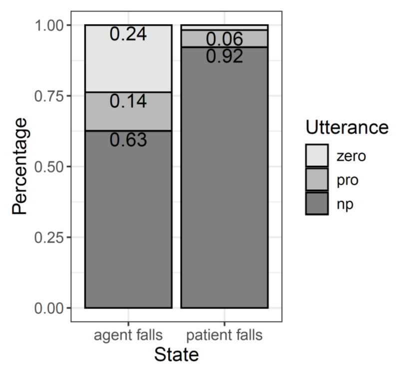




Puzzle

- When the patient falls, speakers mostly use NPs and almost never pronouns to refer to it (92% vs. 6%)[3].
- Then we can expect pragmatic listeners to interpret pronouns almost exclusively as referring to the agent.
- But this was not the case.
- Pronouns were taken to refer to the patient in 33% of cases, which is rather high rate.
- Is this a problem for the Bayesian account of language?





Vanilla RSA[4]

- Pragmatic listener L_1 reasons about a pragmatic speaker S_1, who in turn reasons about a literal listener L_0.
- Set of communicable world states s
- Set of possible utterances u
- Prior probabilities $P(s)$ for each world state
- Meaning function $[[u]](s)$ maps utterances to world states in which they are true

$$P_{L_0}(s|u) \propto P(s) \cdot [[u]](s)$$
$$P_{S_1}(u|s) \propto \exp(\alpha \cdot U(u,s))$$
$$U(u,s) = log P_{L_0}(s|u) - C(u)$$
$$P_{L_1}(s|u) \propto P(s) \cdot P_{S_1}(u|s)$$

- L_0 interprets utterances literally, based on their priors
- L_1 considers the likelihood S_1 would produce a given utterance to convey a particular world state.
- S_1 maximize utility $U(u,s)$ by balancing the informativeness for L_0 against the production cost $C(u)$.
- Parameter α scales the utility, modulating how likely the utterance with the maximum utility is to be chosen.

Noisy-channel RSA[5]

- The noisy-channel extension accounts for the possibility that an utterance may be misperceived.
- Listeners L_0 and L_1 infer the intended utterance u_i based on the perceived utterance u_p
- Speaker S_1 considers that the intended utterance u_i might be misheard as u_p.
- The noise function $P(u_i, u_p)$ specifies the probability that utterance u_i is (mis)perceived as u_p.

$$P_{L_0}(s|u_p) \propto P(s) \cdot \sum_i P(u_i, u_p) \cdot [[u_i]](s)$$
$$P_{S_1}(u_i|s) \propto exp(\alpha \cdot U(u_i, s))$$
$$U(u_i, s) = \sum_p P(u_i, u_p) \cdot log P_{L_0}(s|u_p) - C(u_i)))$$
$$P_{L_1}(s|u_p) \propto P(s) \cdot \sum_i P(u_i, u_p) \cdot P_{S_1}(u_i|s)$$

Our model

- Based on the noisy-channel RSA model.
- We ignore utterance cost $C(u)$ for simplicity.
- Two world states: *agent falls* and *patient falls*.
- Four utterances: *agent*, *patient*, *pro* and *zero*.
- Utterance *pro* is compatible with both world states.
- Other utterances are compatible with only one state.

$$P_{L_0}(s|u_p) \propto P(s) \cdot P_{sal}(s) \cdot \sum_i P(u_i, u_p) \cdot [[u_i]](s)$$
$$P_{S_1}(u_i|s) \propto exp(\alpha \cdot \sum_p P(u_i, u_p) \cdot log P_{L_0}(s|u_p))$$
$$P_{L_1}(s|u_p) \propto P(s) \cdot P_{sal}(s) \cdot \sum_i P(u_i, u_p) \cdot P_{S_1}(u_i|s)$$

- World state priors $P(s)$. Mean from the experiment: $P(agent\ falls) = 0.32, P(patinet\ falls) = 0.68$.
- Salience priors $P_{sal}(s)$ – how plausible is that the agent or the patient will be mentioned next. Parametrized: $P_{sal}(agent\ falls) = ab, P_{sal}(patinet\ falls) = 1 - ab$.
- *Pro* and *zero* can be confused with a parametrized error rates: $P(pro, zero) = e_{pro}, P(pro, pro) = 1 - e_{pro}$, $P(zero, pro) = e_{zero}$ and $P(zero, zero) = 1 - e_{zero}$.
- Noun phrases *agent* and *patient* are always interpreted correctly: $P(agent, agent) = P(patient, patient) = 1$.

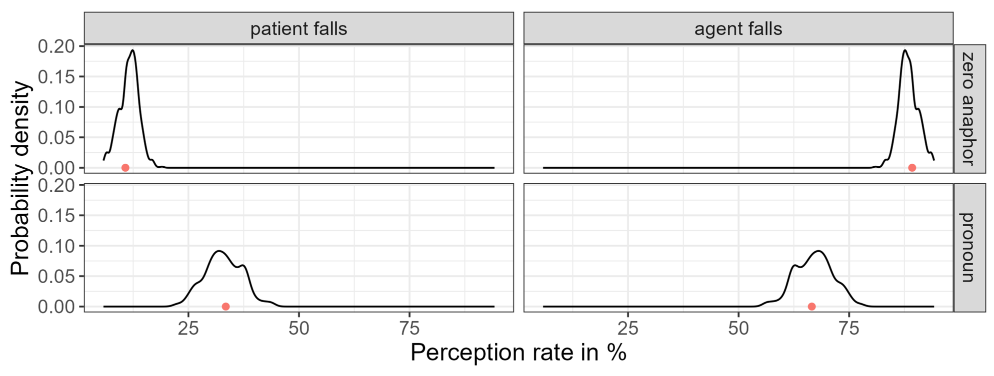
Parameter estimation

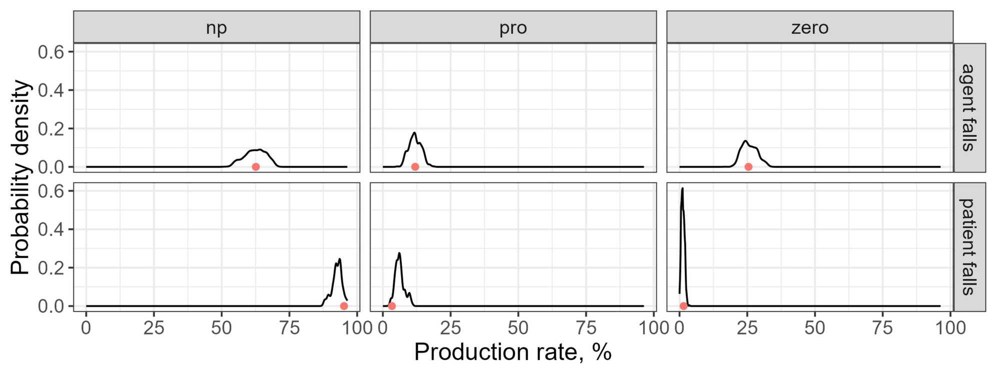
- Bayesian data analysis in WebPPL[6] using the Markov chain Monte Carlo (MCMC) method.
- Mean estimates of the parameter values:
- $\alpha = 3.46,\ ab = 0.63,\ e_{pro} = 0.31,\ e_{zero} = 0.1$

Solution to the puzzle

- The key: speakers also rarely use pronouns when referring to the agent
- Pronoun omissions are more frequent than insertions
- This further reduces the rate of pronouns perceived
- Two types of priors almost cancel each other (63% agent bias vs. 68% patient bias)
- Thus, the rate of a pronoun interpretation towards a world state is roughly proportional to the pronoun generation rate for that state (in accordance with RSA).

Posterior predictive checks





- Posterior predictions of the model.
- Production at the top, perception at the bottom.
- Red dots show the observed rates for each condition.
- Black curves – the model's predicted probability distributions of these rates.
- The red dots lie within the plausible regions of the model predictions, so the model fits the data well.

Conclusions

- We built an RSA model that fits experimental data well.
- Our model provides further evidence that event predictability affects referential expression production and comprehension.
- It further supports the observation that listeners and speakers take the possibility of noise into account.
- We also explained an apparent discrepancy in the experimental data and showed that this is not a counterexample to Bayesian account of conversation.
- Our findings highlight the value of probabilistic models like RSA for capturing pragmatic effects and bridging experimental data with formal theories of language use.

References

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Information

Have a question?
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